

# THE IRRATIONALITY MEASURE OF THE LOGARITHM OF A RATIONAL NUMBER IS 2

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**ABSTRACT.** For every positive rational  $\alpha \neq 1$  the irrationality measure of  $\log \alpha$  equals 2: for every  $\nu > 2$ ,  $|\log \alpha - p/q| \geq q^{-\nu}$  for all integers  $p, q$  with  $q$  sufficiently large. In particular  $\mu(\log 2) = 2$ , where the previous record was 3.5746. The proof is the interpolation-determinant argument that gives  $\mu(\pi) = 2$ , with the  $K$  interpolation centres moved from  $(1, 2\pi i j)$  to  $(\alpha^j, j \log \alpha)$  on the exponential curve  $z \mapsto (e^z, z)$ . The single new ingredient is a version of the separated-weight interpolation theorem with centres in distinct fibres of the exponential coordinate; the arithmetic lower bound pays the height of  $\alpha^j$  in an error term that the existing parameter choice already drives to zero. For real algebraic  $\alpha$  of degree  $D$  the same argument gives  $\mu(\log \alpha) \leq 2D$ , and we explain why ratios of logarithms are not reached. The proof is formalized in Lean 4 over Mathlib, by modifying the formalization of the  $\pi$  theorem module by module.

## 1. INTRODUCTION

For a real number  $\omega$  the irrationality measure  $\mu(\omega)$  is the supremum of the  $\nu$  for which  $|\omega - p/q| < q^{-\nu}$  has infinitely many rational solutions. Dirichlet gives  $\mu \geq 2$  for irrational  $\omega$ , and  $\mu = 2$  for almost every real number. For logarithms of rational numbers the known bounds come from Padé-type constructions:  $\mu(\log 2) \leq 3.5746$  (Marcovecchio [3], improving Rukhadze [6] and Hata [2]), and effective bounds of Baker type  $\mu(\log \alpha) < \infty$  for every algebraic  $\alpha \neq 0, 1$  [1].

The paper [4] proves  $\mu(\pi) = 2$  by a Roth-type argument on the exponential curve: an interpolation determinant whose rows are jets at  $K$  centres  $(1, j\omega)$ ,  $\omega = 2\pi i$ , of polynomials restricted to the curve  $z \mapsto (e^z, z + u)$ , whose columns are weighted monomials, whose arithmetic size is bounded below because the centres are Gaussian integer points after clearing the denominators  $q_i$  of the approximations, and whose analytic size is bounded above because rows at different centres test the same entire functions and collide. The theorem is machine-verified in Lean 4 over Mathlib, and so is Theorem A below (Section 6). We cite [4] by its section, equation and result numbers throughout.

**Theorem A.** *Let  $\alpha$  be a positive rational number,  $\alpha \neq 1$ , and  $\omega = \log \alpha$ . For every real  $\nu > 2$  there is  $Q(\nu, \alpha)$  such that*

$$\left| \omega - \frac{p}{q} \right| \geq q^{-\nu} \quad \text{for all integers } p, q \text{ with } q \geq Q(\nu, \alpha).$$

Hence  $\mu(\log \alpha) = 2$ ; in particular  $\mu(\log 2) = 2$ .

The argument of [4] uses one property of  $\pi$ :  $e^{2\pi i j} = 1$ , which places the centre  $(1, j\omega)$  on the curve at  $z = j\omega$  (Lemma 3.2 there). For  $\omega = \log \alpha$  the analogous fact is  $e^{j\omega} = \alpha^j$ , which places the centre  $(\alpha^j, j\omega)$  on the curve. The centres therefore lie in distinct fibres of the exponential coordinate  $Y$ , and the interpolation theorem of [4], stated for centres with  $Y = 1$ , must be restated for centres  $(y_j, c_j)$  with distinct  $y_j$ . Section 2 does this; the proof is that of [4], Section 2, with three local changes which we display. Section 3 carries the  $Y$ -height of  $\alpha^j$  through the determinant estimates: it costs  $(K - 1) \log b/w_0$  in the arithmetic lower bound ( $\alpha = a/b$ ) and replaces the radius  $100K$  by  $100K(1 + |\log \alpha|)$  in the collision estimate, both of the form  $O_\alpha(K/w_0)$ , which the parameter choice of [4], Section 4, sends to zero. For

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$\alpha = 2$  the arithmetic lower bound is literally Lemma 3.1 of [4]. Section 5 records what the method gives for algebraic  $\alpha$  and why it gives nothing for  $\log 3/\log 2$ .

**Notation.** Coordinates  $Y, X_1, \dots, X_m$ ; a positive rational weight vector  $W = (w_0, \dots, w_m)$  gives the  $W$ -degree  $w_0 h + \sum w_i \alpha_i$  of  $Y^h X^\alpha$ , and  $P_W(H)$  is the space of polynomials of  $W$ -degree at most  $H$ . For  $V = (v_0, \dots, v_m)$ ,  $J_V(H)$  is the quotient of  $\mathbb{C}[[t, u_1, \dots, u_m]]$  by the series all of whose monomials have  $V$ -weight at least  $H$ . We write  $w\beta = \sum_i w_i \beta_i$ ,  $w_* = \min_i w_i$ ,  $\Lambda = 4 \log 2$ , and all logarithms are natural.

## 2. INTERPOLATION WITH CENTRES IN DISTINCT FIBRES

**Theorem 2.1** (Separated-weight interpolation, general fibres). *Fix integers  $m, K \geq 1$  and positive rationals  $w_0, v_0, \theta$  with  $0 < \theta < 1$ ,*

$$(1) \quad K\theta^m < 1, \quad K \frac{w_0}{v_0} \theta^m < 1.$$

*There are successive lower thresholds for positive rationals  $w_1, \dots, w_m$ , each depending only on these data and the previously chosen weights, with the following property. Put  $W = (w_0, \dots, w_m)$ ,  $V = (v_0, w_1/\theta, \dots, w_m/\theta)$ . For any points*

$$a_j = (y_j, c_{j1}, \dots, c_{jm}) \in \mathbb{C}^\times \times \mathbb{C}^m, \quad 0 \leq j < K,$$

*such that whenever  $y_j = y_{j'}$  with  $j \neq j'$  the coordinates  $c_{ji}, c_{j'i}$  differ for every  $i$ , the map*

$$(2) \quad P_W(H) \longrightarrow \bigoplus_{j=0}^{K-1} J_V(H), \quad P \longmapsto \left( P(y_j(1+t), (c_{ji} + u_i + \log(1+t))_{i=1}^m) \right)_{j=0}^{K-1},$$

*is surjective for all sufficiently large, sufficiently divisible  $H$ . The divisibility can be chosen using only the weights; the remaining threshold for  $H$  may depend on the points.*

For  $y_j = 1$  this is Theorem 2.1 of [4]. When the  $y_j$  are pairwise distinct the condition on the  $c_{ji}$  is empty.

*Proof.* We follow [4], Section 2, and indicate the changes. The weights are chosen as in (2.7)–(2.9) there:  $\sigma > 0$  small with  $(1 + 3\sigma)^{m+1} K(w_0/v_0)\theta^m < 1$ ,  $(1 + 3\sigma)^m K\theta^m < 1$ ,  $(1 + \sigma)\theta < 1$ ;  $v_i = w_i/\theta$ ;  $L_0 = m + 2$ ;  $C_*$  as in (2.8); and  $w_1, \dots, w_m$  successively so large that  $\prod_{a \in A} w_a > C_* \prod_{b \in B} v_b$  for all index sets  $A, B$  of equal size whose largest differing index lies in  $A$ . This involves no point.

Let  $C$  be an irreducible curve with generic point in the affine chart, with smooth projective model,  $W$ -degree

$$\deg_W C = \sum_P \max\left(0, -\frac{\text{ord}_P Y}{w_0}, -\frac{\text{ord}_P X_1}{w_1}, \dots, -\frac{\text{ord}_P X_m}{w_m}\right)$$

as in (2.10), and, at a branch  $P$  through  $a_j$ ,

$$(3) \quad h_P = \min\left(\frac{\text{ord}_P(Y - y_j)}{v_0}, \frac{\text{ord}_P(X_1 - c_{j1} - \log(Y/y_j))}{v_1}, \dots, \frac{\text{ord}_P(X_m - c_{jm} - \log(Y/y_j))}{v_m}\right),$$

where  $\log(Y/y_j)$  is its formal expansion at  $Y = y_j$ , well defined along the branch since  $Y - y_j$  has positive order there. This replaces (2.11), where  $y_j = 1$ . In the local coordinate  $t = Y/y_j - 1$  at  $a_j$  the jet (2) is  $P(y_j(1+t), c_j + u + \log(1+t))$ , so  $h_P$  measures, in  $V$ -weight, the contact of the branch with the jet. The frame  $D_0 = Y\partial_Y + \sum_i \partial_{X_i}$ ,  $D_i = \partial_{X_i}$  of [4] is unchanged. In the coordinates  $(t, u)$  at  $a_j$ , where  $Y = y_j(1+t)$  and  $X_i = c_{ji} + u_i + \log(1+t)$ , the chain rule gives  $\partial_t = y_j\partial_Y + (1+t)^{-1} \sum_i \partial_{X_i}$ , hence  $(1+t)\partial_t = D_0$  and  $\partial_{u_i} = D_i$ : the action of the frame on jets is the same at every centre, which is all that the order bookkeeping of [4] (the display after (2.13), “each monomial has order at least its weight times  $h_P$ ”) uses.

*Curve inequality.* We claim

$$(4) \quad \deg_W C \geq (1 + \sigma) \sum_{P \mapsto a_j} h_P,$$

the sum over all branches through prescribed points. Suppose it fails. The construction of the auxiliary polynomial  $F_N$  (of  $W$ -degree at most  $N$ ,  $V$ -order at least  $(1 + 3\sigma)N$  at every centre) is a count of coefficients against jet conditions; the number of jet conditions at  $a_j$  is the number of monomials  $t^s u^\beta$  of  $V$ -weight below  $(1 + 3\sigma)N$ , independent of  $y_j$ , so the count (2.13) is unchanged. The zero–pole comparison on  $C$  (poles of  $F_N|_C$  of total degree at most  $N \deg_W C$ , zeros of order at least  $(1 + 3\sigma)N h_P$  at the branches through centres) uses only (3) and the order bookkeeping, and gives  $F_N|_C = 0$  together with the vanishing (2.14) of all frame derivatives of cost at most  $\sigma N$  on  $C^\circ$ . The persistent component  $Z \supseteq C^\circ$ , the normal-basis inequality (2.15) from Lemma 2.2 there, the weight separation, and the residue argument are statements about  $Z$ ,  $F_N$  and the frame, and involve no centre. They give:  $Y$  is constant on  $Z$ .

Here is the first change. In [4] one concludes “ $Y = 1$  on  $C^\circ$ ” because  $C$  meets a centre with  $Y = 1$ . Now  $C^\circ \subseteq Z$  meets some centre  $a_{j_0}$  (otherwise there is nothing to prove), so  $Y \equiv y_{j_0}$  on  $C$ . Consequently every centre that  $C$  meets lies in the fibre  $Y = y_{j_0}$ , and the sum in (4) runs over branches through centres of that fibre only.

*The fibre argument.* Work in the affine fibre  $Y = y_{j_0}$ , in which  $h_P = \min_i \text{ord}_P(X_i - c_{ji})/v_i$  for the centres  $a_j$  of the fibre (the  $Y$ -term of (3) is  $+\infty$  and  $\log(Y/y_j) = 0$  there). This is the second change: in [4] all  $K$  centres lie in the fibre, here only those with  $y_j = y_{j_0}$ , say  $K' \leq K$  of them. The auxiliary polynomial  $G_N(X_1, \dots, X_m)$  of weighted degree at most  $N$  with order at least  $(1 + 3\sigma)N$  at these  $K'$  centres exists because the ratio of conditions to coefficients tends to  $K'(1 + 3\sigma)^m \theta^m \leq K(1 + 3\sigma)^m \theta^m < 1$ . The zero–pole comparison, the chain  $U_r$ , the persistent component  $Z'$  of codimension  $1 \leq k \leq m - 1$  (for  $m \geq 2$ ; for  $m = 1$  the contradiction is immediate), and the unique normal index set are as in [4]; they give that some  $X_i$  is constant on  $Z'$ .

Third change: “coordinatewise distinctness shows  $Z'$  meets at most one prescribed point” is the hypothesis of Theorem 2.1 restricted to the fibre: centres sharing  $y_{j_0}$  have distinct  $i$ -th coordinates. When the  $y_j$  are pairwise distinct,  $K' = 1$ , the fibre has a single centre, and this second stage (the polynomial  $G_N$ , the chain  $U_r$ , the component  $Z'$ ) is not needed at all. In either case  $C$  meets only one centre  $a_j$  of the fibre, and with a coordinate  $X_i$  nonconstant on  $C$ ,

$$\deg_W C \geq \theta^{-1} \sum_{P \mapsto a_j} h_P > (1 + \sigma) \sum_{P \mapsto a_j} h_P$$

by  $(1 + \sigma)\theta < 1$ , contradicting the failure of (4).

*From the curve inequality to jets.* Section 2.3 of [4] works on the weighted projective closure with the weighted point ideals  $I_j = (t^{R/v_0}, (u_i')^{R/v_i})$  at the centres, the blow-up along them, the nefness of  $A - (1 + \sigma)E$  (from (4) applied to every curve, including curves missing the chart, which are handled there without reference to the centres), ampleness of  $A - E$ , and Serre vanishing giving  $H^1 = 0$  and surjectivity onto the truncated jets; the final step undoes the truncation by the invertible filtered substitution. The only point-dependent object is the local coordinate  $t = Y - 1$  at the centre, which becomes  $t = Y/y_j - 1 = (Y - y_j)/y_j$ , a regular parameter at  $a_j$  since  $y_j \neq 0$ ; (2.18)–(2.23) there are verbatim. The thresholds for  $w_1, \dots, w_m$  come from (2.7) and (2.9) and involve no point; the threshold for  $H$  (through the choice of  $N$  and the vanishing) may depend on the points.  $\square$

### 3. THE DETERMINANT

Let  $\alpha = a/b$  with coprime positive integers  $a \neq b$ ,  $\omega = \log \alpha \neq 0$ . Let  $\nu > 2$  and suppose integers  $p_i, q_i \geq 2$  satisfy

$$(5) \quad \left| \omega - \frac{p_i}{q_i} \right| \leq q_i^{-\nu}, \quad w_i = \lceil \log q_i \rceil, \quad r_i = \frac{p_i}{q_i} \quad (1 \leq i \leq m),$$

with  $m, K, w_0, v_0, \theta, w_1, \dots, w_m$  admissible for Theorem 2.1. Since  $\omega \neq 0$ ,  $p_i \neq 0$  once  $q_i$  is large. Fix  $F_0 > 2/\theta$ ,  $T_i = F_0 w_i / v_0$ , and the truncated logarithms  $G_i(t) = \sum_{1 \leq k < T_i} (-1)^{k+1} t^k / k$  of (3.2) in [4]; replacing  $\log(1+t)$  by  $G_i$  is an invertible filtration-preserving substitution, so Theorem 2.1 remains surjective with the  $G_i$  (Section 3 of [4]).

Columns are the monomials  $Y^h X^\alpha$  with  $w_0 h + w\alpha \leq H$ ; rows are the triples  $\rho = (j, s, \beta)$  with  $0 \leq j < K$ ,  $v_0 s + w\beta/\theta < H$ ; the entry is the coefficient of  $t^s u^\beta$  in

$$(6) \quad P(\alpha^j(1+t), jr_1 + G_1(t) + u_1, \dots, jr_m + G_m(t) + u_m).$$

By Theorem 2.1 with centres  $(\alpha^j, jr_1, \dots, jr_m)$ , whose  $Y$ -coordinates  $\alpha^j$  are pairwise distinct because  $\alpha > 0, \alpha \neq 1$ , for all large divisible  $H$  there is a nonzero square minor  $\Delta_H$  using every row. Its size  $M = M_H$  satisfies (3.5) of [4],  $M \sim K\theta^m H^{m+1} / ((m+1)! v_0 \prod_i w_i)$ , and  $\bar{b} = \bar{b}_H = (MH)^{-1} \sum_\rho w\beta \in [0, \theta]$ .

**Lemma 3.1** (Arithmetic lower bound). *With  $E_{\text{ar}} = \Lambda F_0 m / v_0 + \Lambda \sum_i 1/w_i + \theta/w_*$  as in (3.6) of [4] and*

$$E'_{\text{ar}} = E_{\text{ar}} + \frac{(K-1) \log b}{w_0},$$

*every minor chosen as above satisfies  $\log |\Delta_H| / (MH) \geq -(1 - \bar{b}) - E'_{\text{ar}}$ .*

*Proof.* For  $P = Y^h X^\alpha$  the entry of row  $(j, s, \beta)$  equals

$$\alpha^{jh} \prod_i \binom{\alpha_i}{\beta_i} [t^s] (1+t)^h \prod_i (jr_i + G_i(t))^{\alpha_i - \beta_i}$$

when  $\alpha \geq \beta$  and 0 otherwise. Scale column  $Y^h X^\alpha$  by  $\prod_i q_i^{\alpha_i}$ , row  $(j, s, \beta)$  by  $\prod_i q_i^{-\beta_i}$ , and multiply by  $D_H = \prod_i \text{lcm}(1, \dots, T_i - 1)^{\lfloor H/w_i \rfloor}$ , which clears the denominators of the  $G_i$  (proof of Lemma 3.1 in [4]); since  $q_i jr_i = jp_i \in \mathbb{Z}$ , all factors except  $\alpha^{jh}$  are then integers. Put  $h_{\max} = \lfloor H/w_0 \rfloor$  and scale row  $(j, s, \beta)$  additionally by  $b^{jh_{\max}}$ , turning  $\alpha^{jh}$  into  $a^{jh} b^{j(h_{\max} - h)} \in \mathbb{Z}$ . The scaled matrix is over  $\mathbb{Z}$  with nonzero determinant, so its determinant has modulus at least 1. The row scalings by  $b^{jh_{\max}}$  contribute at most  $\sum_\rho j h_{\max} \log b \leq M(K-1)(H/w_0) \log b$  to the logarithm; the remaining scalings are accounted for exactly as in (3.8) of [4]:  $\sum_{\text{cols}} \sum_i \alpha_i \log q_i \leq MH$ ,  $\sum_{\text{rows}} \sum_i \beta_i \log q_i \geq MH\bar{b} - MH\theta/w_*$ , and  $\log D_H \leq MH(\Lambda F_0 m / v_0 + \Lambda \sum_i 1/w_i)$ .  $\square$

For  $\alpha = 2$  one has  $b = 1$  and Lemma 3.1 is Lemma 3.1 of [4] with  $\mathbb{Z}$  in place of  $\mathbb{Z}[i]$ .

**Lemma 3.2** (Exact row translation). *Put  $z = j\omega + \log(1+t)$ . Then  $e^z = \alpha^j(1+t) = Y$  and  $X_i = z + u_i + \varepsilon_i + \tau_i(t)$  with  $\varepsilon_i = j(r_i - \omega)$ ,  $\tau_i = G_i - \log(1+t)$ . Every row of (6) is a sum of at most  $Q_H = (\lfloor H \rfloor + 1)^{2m} (\lfloor H/v_0 \rfloor + 1)$  scalar multiples of the test rows  $([t^\ell] f_{a,P}(j\omega + \log(1+t)))_P$ ,  $a \geq \beta$ ,  $wa \leq H$ ,  $0 \leq \ell \leq s$ , where*

$$f_{a,P}(z) = [u^a] P(e^z, z + u_1, \dots, z + u_m)$$

*is an entire function of  $z$  independent of  $j$ , and a scalar with index  $a$  satisfies  $|\xi| \leq \exp\{-\nu w(a - \beta) + HE_{\text{tr}}\}$  with  $E_{\text{tr}} = \nu/F_0 + \log 2/v_0 + (\log 4 + \log(2K) + \nu)/w_*$  as in (3.12) of [4].*

*Proof.* The identity  $e^{j\omega} = \alpha^j$  gives  $e^z = Y$ ; the rest is the substitution of Lemma 3.2 in [4], and the identity (3.13) and its bounds are verbatim, with  $|\varepsilon_i| \leq (K-1)q_i^{-\nu}$ , which is smaller than the bound  $2Ke^\nu e^{-\nu w_i}$  used there.  $\square$

**Lemma 3.3** (Collision estimate). *Put  $R_\alpha = 100K(1 + |\log \alpha|)$ ,  $c = (\log 2)/4$  and*

$$E'_{\text{hol}} = \frac{R_\alpha}{w_0} + \frac{\log 2}{v_0} + \frac{\log(2R_\alpha)}{w_*}.$$

*Choose one translated test row from each row expansion of Lemma 3.2, remove the scalars, and let  $T$  be the resulting matrix and  $n_a$  the number of rows with transverse index  $a$ . Then  $|\det T| \leq \exp\{-c \sum_a n_a^2 + MH(E'_{\text{hol}} + \rho_H)\}$  with  $\rho_H \rightarrow 0$  independent of the choices.*

*Proof.* On  $|z| \leq R_\alpha$  one has  $|f_{a,P}(z)| \leq \exp\{H(R_\alpha/w_0 + \log(2R_\alpha)/w_*)\}$  by (3.10) of [4], since  $h \leq H/w_0$  and  $|\alpha| \leq H/w_*$ . On  $|t| = 1/2$ ,  $|j\omega + \log(1+t)| \leq (K-1)|\log \alpha| + \log 2 \leq R_\alpha/2$ . The expansion of each  $f_{a,P}$  in powers of  $z/R_\alpha$ , the vanishing of determinant terms with a repeated pair  $(a, d)$ , the bound  $\sum_\rho d_\rho \geq \sum_a \binom{n_a}{2}$ , and the geometric decay are those of Lemma 3.3 in [4] with  $R$  replaced by  $R_\alpha$ .  $\square$

**Proposition 3.4** (Comparison). *Let  $0 < \eta < 1$ ,  $A \in (0, 1)$ ,  $L = \eta^2 K \theta^m / ((m+1)v_0 A^m)$ ,  $E'_{\text{an}} = E'_{\text{tr}} + E'_{\text{hol}}$ , and*

$$g = \nu(A(1-\eta) - \theta) - (1-\theta).$$

*If  $g > 0$ ,  $E'_{\text{ar}} + E'_{\text{an}} < g$  and  $cL > 1 + E'_{\text{ar}} + E'_{\text{an}}$ , then for all sufficiently large divisible  $H$  the bounds of Lemma 3.1 and Lemmas 3.2–3.3 are incompatible.*

*Proof.* Proposition 3.4 of [4], whose proof uses only the three lemmas, the simplex count  $D_A(H) \sim (AH)^m / (m! \prod w_i)$  of rows with  $wa \leq AH$ , and  $\bar{b} \leq \theta$ : either at least  $\eta M$  rows have  $wa \leq AH$ , and Cauchy–Schwarz gives  $\sum_a n_a^2 \geq MH(L - o(1))$ , so Lemma 3.3 beats Lemma 3.1 when  $cL > 1 + E'_{\text{ar}} + E'_{\text{an}}$ ; or more than  $(1-\eta)M$  rows have  $wa > AH$ , and the scalars of Lemma 3.2 supply  $\exp\{-\nu(A(1-\eta) - \bar{b})MH\}$ , which beats Lemma 3.1 when  $\nu(A(1-\eta) - \bar{b}) - (1-\bar{b}) \geq g > E'_{\text{ar}} + E'_{\text{an}}$ , the minimum over  $\bar{b} \in [0, \theta]$  being at  $\bar{b} = \theta$  since  $\nu > 1$ .  $\square$

#### 4. PROOF OF THEOREM A

**Lemma 4.1** (Parameters; Lemma 4.1 of [4]). *Let  $\nu > 2$ . There are rationals  $0 < \theta < A < B < 1$  and  $C > 1$  with*

$$\nu(A - \theta) > 1 - \theta, \quad A^2 < \theta, \quad B < 1/C, \quad B < C\theta.$$

*Proof.* Take  $b \in (1/2, 1 - 1/\nu)$ ,  $\theta = 1 - \delta$ ,  $A = 1 - b\delta$  with  $\delta$  small: then  $\nu(A - \theta) - (1 - \theta) = (\nu(1 - b) - 1)\delta > 0$  and  $\theta - A^2 = (2b - 1)\delta - b^2\delta^2 > 0$ ; since  $A < \sqrt{\theta}$  one has  $A/\theta < 1/A$ , so  $C \in (A/\theta, 1/A)$  and then  $B \in (A, \min(1/C, C\theta))$  exist.  $\square$

*Proof of Theorem A.* Fix  $\nu > 2$  and suppose, for contradiction, that arbitrarily large  $q$  admit  $p$  with  $|\omega - p/q| \leq q^{-\nu}$ . Take  $\theta, A, B, C$  from Lemma 4.1, then  $\eta > 0$  so small that  $g = \nu(A(1-\eta) - \theta) - (1-\theta) > 0$ , put  $\varepsilon_0 = \min(g/2, 1/2)$ , and fix  $F_0 > 2/\theta$  with  $\nu/F_0 < \varepsilon_0/3$ . For an integer  $m$  put

$$K = \lfloor C^m \rfloor, \quad w_0 = B^{-m}, \quad v_0 = 2K\theta^m w_0.$$

Then  $K\theta^m < 1$  and  $K(w_0/v_0)\theta^m = 1/2$ , so (1) holds;  $K/w_0 \leq (CB)^m \rightarrow 0$ ;  $v_0 \sim 2(C\theta/B)^m \rightarrow \infty$ ; and  $L = \eta^2(B/A)^m/(2(m+1)) \rightarrow \infty$  as  $m \rightarrow \infty$ . Choose  $m$  so large that

$$(7) \quad \frac{\Lambda F_0 m + 2 \log 2}{v_0} + \frac{100K(1 + |\log \alpha|) + (K-1) \log b}{w_0} < \frac{\varepsilon_0}{3}, \quad cL > 2;$$

this is possible because the second summand is at most  $(CB)^m(100(1 + |\log \alpha|) + \log b)$ . Fix  $m, K, w_0, v_0$ , hence the  $Y$ -coordinates  $\alpha^j$  of the centres. Choose approximations  $p_i/q_i$  satisfying (5),  $w_i = \lfloor \log q_i \rfloor$ , successively so large that they meet the thresholds of Theorem 2.1 (which depend only on the fixed data and the earlier  $w$ 's) and

$$(8) \quad \Lambda \sum_i \frac{1}{w_i} + \frac{\theta + \log 4 + \log(2K) + \nu + \log(2R_\alpha)}{w_*} < \frac{\varepsilon_0}{3}.$$

Then  $E'_{\text{ar}} + E'_{\text{an}} < \varepsilon_0 \leq g/2$  and  $cL > 2 > 1 + E'_{\text{ar}} + E'_{\text{an}}$ , and Proposition 3.4 gives a contradiction for large  $H$ . Hence  $|\omega - p/q| > q^{-\nu}$  for all  $q \geq Q(\nu, \alpha)$ . Taking  $\nu = 3$  shows that  $\omega$  is not rational (a rational  $\omega = p/q$  would give infinitely many exact representations  $p'/q'$ ), so  $\mu(\omega) \geq 2$  by Dirichlet, and  $\mu(\omega) = 2$ .  $\square$

The threshold  $Q(\nu, \alpha)$  is ineffective, exactly as in [4]: the dimension  $m$  and the approximations are chosen against a hypothetical sequence of exceptionally good approximations.

**Corollary 4.2.** *For every positive rational  $\alpha \neq 1$  and every  $\nu > 2$ ,  $\|q \log \alpha\| \geq q^{1-\nu}$  for all large  $q$ , where  $\|\cdot\|$  is the distance to the nearest integer; equivalently  $|\alpha^q - e^p| \geq \frac{1}{2}e^p q^{1-\nu}$  for all large  $q$  and all  $p$ .*

## 5. ALGEBRAIC $\alpha$ , AND THE LIMITS OF THE METHOD

**Proposition 5.1.** *Let  $\alpha > 0$  be a real algebraic number of degree  $D$ ,  $\alpha \neq 1$ . Then  $\mu(\log \alpha) \leq 2D$ .*

*Proof sketch.* Run Sections 3–4 with centres  $(\alpha^j, jr_i)$ . After the column and row scalings and an extra row scaling by  $d^{jh_{\max}}$ , where  $d$  is a denominator of  $\alpha$ , the matrix has entries in the ring of integers of  $\mathbb{Q}(\alpha)$ , and  $\Delta' = (\text{scalings}) \cdot \Delta_H$  is a nonzero algebraic integer, so  $|N_{\mathbb{Q}(\alpha)/\mathbb{Q}}(\Delta')| \geq 1$ , i.e.  $\log |\Delta_H| \geq -DS - \sum_{\sigma \neq \text{id}} \log |\sigma \Delta_H|$  with  $S = MH(1 - \bar{b}) + MHE'_{\text{ar}}$ . The conjugate  $\sigma \Delta_H$  is the determinant of the same matrix with  $\alpha$  replaced by  $\sigma(\alpha)$ ; its rows are jets of  $P(\lambda^j e^z, z + u + \varepsilon + \tau)$  with  $\lambda = \sigma(\alpha)/\alpha$  and  $z = j\omega + \log(1 + t)$ , so the column  $Y^h X^\alpha$  tests  $\lambda^{jh} f_{a,P}(z)$ : the factor  $\lambda^{jh}$  depends on the row through  $j$  and on the column through  $h$ , and rows at different centres with the same  $(a, d)$  are no longer proportional. Only Hadamard's bound  $\log |\sigma \Delta_H| \leq MH \cdot o(1)$  is available for the conjugates. The comparison then needs  $\nu(A(1 - \eta) - \theta) - D(1 - \theta) > 0$ , and with  $\theta = 1 - \delta$ ,  $A = 1 - b\delta$ ,  $b > 1/2$  this is  $\nu(1 - b) > D$ , giving every  $\nu > 2D$ .  $\square$

If  $\lambda$  were a root of unity of order  $n$ , grouping the rows by  $j \bmod n$  would restore the collision at the cost  $L \rightarrow L/n$ ; but  $\sigma(\alpha) = \zeta \alpha$  for all  $\sigma$  forces  $\alpha^n \in \mathbb{Q}$ , which is the rational case.

**Remark 5.2** (Ratios of logarithms). *The method does not reach  $\xi = \log 3 / \log 2$  or any ratio of logarithms, for which the known bounds come from linear independence measures for  $1, \log 2, \log 3$  (Rhin [5]; see Zudilin [7] for the state of the art and the method). An approximation  $p/q$  to  $\xi$  is the statement  $3^q / 2^p = e^\varepsilon$  with  $|\varepsilon| \leq q^{1-\nu} \log 2$ . The method requires, for each approximation, an algebraic centre on the curve  $z \mapsto (e^z, z + u)$  with a rational jet whose arithmetic cost matches the weights:  $X$ -heights  $\log q_i$  against  $X$ -degrees of weight  $w_i = \log q_i$ , and  $O(K)$   $Y$ -heights against  $Y$ -degrees  $h \leq H/w_0$  with  $w_0$  fixed. For  $\log 2$  the algebraic object attached to  $p/q$  is  $p/q$  itself, of height  $\log q$ , and the error  $q^{-\nu}$  is exponentially small in that height. For  $\xi$  the algebraic object is  $3^q / 2^p$ , of height  $\asymp q$ , against the error  $q^{1-\nu}$ , polynomially small in the height: placing it in a centre costs  $2q \log 3 / w_0$  per unit of  $H$ , unbounded, while placing the approximation in the jet (the curve  $Y = 2^X$ , whose tangent field  $Y \partial_Y + (\log 2)^{-1} \partial_X$  is not defined over  $\mathbb{Q}$ ) costs denominators  $q^s$  in the derivative direction, whose weight  $v_0$  is fixed. This is Baker's regime, not the Roth regime of the determinant. The number  $e^\xi$  is transcendental by Gelfond–Schneider, so no centre of the exponential curve attached to  $\xi$  is algebraic.*

**Remark 5.3** (Falsification tests). *The argument must fail for  $\alpha = 1$ : then  $\omega = 0$ ,  $p_i = 0$  for every good approximation, and the centres  $(1, 0)$  coincide, violating the hypothesis of Theorem 2.1; and Theorem A is false for  $\omega = 0$ . It must fail for a general transcendental  $\omega$ , since Liouville numbers exist: for such  $\omega$  the point  $e^{j\omega}$  is not algebraic, and Lemma 3.1 has no integral matrix. For  $\omega = \log \alpha$  with  $\alpha$  algebraic, Baker's theorem gives  $\mu(\omega) < \infty$ , consistent with Proposition 5.1, and  $\log \alpha$  is never a Liouville number.*

## 6. FORMAL VERIFICATION

The directory `lean/` accompanying this paper contains a Lean 4 development over Mathlib (Lean v4.34.1, Mathlib at the commit pinned by [4]), vendored from the formalization of [4] and modified module by module: 863 files, 94,643 lines, with the original  $\pi$  development kept byte-identical alongside it. The file `Main.lean` proves

```
theorem log_irrationalityExponent (α : ℚ) (hα : 0 < α) (hα1 : α ≠ 1) :
  irrationalityExponent (Real.log α) = 2
```

```
theorem log_two_irrationalityExponent : irrationalityExponent (Real.log 2) = 2
```

```
theorem log_not_liouvilleWith (α : ℚ) (hα : 0 < α) (hα1 : α ≠ 1) {p : ℝ} (hp : 2 < p) :
  ¬ LiouvilleWith p (Real.log α)
```

together with `log_main`, the explicit form of Theorem A in the vocabulary of the challenge file of [4]: for every real  $\nu > 2$  an integer  $Q \geq 2$  with  $q^{-\nu} \leq |\log \alpha - p/q|$  for all integers  $p, q$  with  $q \geq Q$ , and the supremum of the set of  $\nu > 0$  admitting infinitely many rationals  $r$  with  $0 < |\log \alpha - r| < \text{den}(r)^{-\nu}$  equal to 2. Here `irrationalityExponent` is the irrationality measure as defined in [4] (that supremum), vendored unchanged under its original name, and `LiouvilleWith` is Mathlib's.

The geometric side (Theorem 2.1) is the development of [4], Section 2, with the base point  $Y = 1$  of the centres replaced by a parameter  $y$  and the  $j$ -th centre placed at  $y^j$ ; the arithmetic side is Lemmas 3.1–3.3, Proposition 3.4 and Section 4 as written, with the row scaling  $b^{j h_{\max}}$ , the identity  $e^{j\omega} = \alpha^j$ , the radius  $100K(1 + |\omega|)$  and the error term  $(K - 1) \log b/w_0$  appearing where the text puts them. The harness `AxiomCheck.lean` checks, with `#guard_msgs`, that the theorems above and the intermediate lemmas depend only on `propext`, `Classical.choice` and `Quot.sound`; there is no `sorry`, no `native_decide` and no added axiom.

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